

Project Sketch for “The Multilingual Hivemind: Output Homogeneity, Model Competence, and Diversity Across the World’s Major Languages”

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Overview

Jiang et al. (2025) studied the “Artificial Hivemind” effect in language models (LMs); on open-ended tasks, LMs generally converge on strikingly similar responses. Their study, built on the Infinity-Chat dataset of 26K real-world queries and spanning 70+ LMs, was conducted entirely in English. This leaves a central question unanswered: is the Hivemind effect a universal property of language models, or is it an artifact of English-centric training and alignment?

We propose a multilingual extension of the Artificial Hivemind study. We investigate output diversity across all languages with more than 100 million total speakers, as listed in Ethnologue (2025): English, Mandarin Chinese, Hindi, Spanish, Modern Standard Arabic, French, Bengali, Portuguese, Indonesian, Urdu, Russian, Standard German, Japanese, Nigerian Pidgin, and Egyptian Arabic. We additionally include Swedish, for which we have native-speaker competence within our team.

Research Questions

1. Does the Artificial Hivemind effect generalize across languages, or does output homogeneity vary with the degree to which a language is represented in LM training and Alignment?
2. Is there a systematic relationship between model competence in a given language (as measured by standard multilingual benchmarks) and the diversity of its open-ended outputs in that language?
3. Are the original paper’s diversity findings robust to the choice of evaluation metric, particularly when applied across typologically diverse languages where English-centric embedding models may behave differently?
4. Is it possible to get a rough overview of the synthetic data landscape that LMs are trained on, and also how this differs from language to language? Does this offer any insight into the results of the Artificial Hivemind effect?

Approach

The resulting set of 16 languages includes several analytically valuable contrasts: Modern Standard Arabic (a purely literary register with zero native speakers) alongside Egyptian Arabic (a spoken dialect with 84 million native speakers); Hindi and Urdu, which are largely mutually intelligible in colloquial registers but differ in script and formal vocabulary; and Nigerian Pidgin, an English-based creole with likely minimal representation in LM training corpora. Chinese, being a pictographic language, may also yield interesting results.

We draw a focused subset of 50–100 open-ended prompts from Infinity-Chat, sampled across the paper’s taxonomy categories (e.g., creative generation, brainstorming, open-ended reasoning).

Prompts are translated into each target language using a strong multilingual LM. Crucially, no existing multilingual dataset fills this role: available parallel resources such as MKQA and MultiQ are designed for factual question answering with single correct answers, making them unsuitable for measuring diversity on open-ended tasks where multiple valid responses exist. Our machine-translated prompt set, while imperfect, addresses a genuine gap in multilingual evaluation infrastructure.

To verify the reliability of our machine translations, we can also use MKQA (Longpre et al., 2021), which provides expert bilingual translations of 10K questions across 26 languages. We additionally validate translations where possible via native speakers within our team.

To measure model competence per language, we draw on existing multilingual benchmarks with expert-quality annotations: MKQA for open-domain factual QA and Belebele for reading comprehension.

Related Work

1. Jiang, L., Chai, Y., Li, M., Liu, M., Fok, R., Dziri, N., ... & Choi, Y. (2025). Artificial hivemind: The open-ended homogeneity of language models (and beyond). arXiv preprint arXiv:2510.22954.
2. Longpre, S., Lu, Y., & Daiber, J. (2021). MKQA: A linguistically diverse benchmark for multilingual open domain question answering. *Transactions of the Association for Computational Linguistics*, 9, 1389-1406.
3. Bandarkar, L., Liang, D., Muller, B., Artetxe, M., Shukla, S. N., Husa, D., ... & Khabsa, M. (2024, August). The belebele benchmark: a parallel reading comprehension dataset in 122 language variants. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 749-775).
4. Wang, Q., Pan, S., Linzen, T., & Black, E. (2025, November). Multilingual prompting for improving llm generation diversity. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing* (pp. 6378-6400).
5. Shaib, C., Govindarajan, V. S., Barrow, J., Sun, J., Siu, A., Wallace, B. C., & Nenkova, A. (2025, December). Standardizing the Measurement of Text Diversity: A Tool and Comparative Analysis. In *Proceedings of The 14th International Joint Conference on Natural Language Processing and The 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics: System Demonstrations* (pp. 36-46).

Datasets

1. Infinity-Chat (Jiang et al., 2025): 26K open-ended queries with taxonomy annotations, available via HuggingFace and used by the Hivemind paper. We use a curated subset of 50-100 prompts alongside our multilingual translations as the diversity evaluation set.
2. MKQA (Longpre et al., 2021): 10K factual QA pairs with expert translations in 26 languages. Used in a dual role: (1) as a benchmark for validating the quality of our machine translation pipeline, and (2) as a task performance benchmark for measuring model competence per language.
3. Belebele (Bandarkar et al., 2024): Professionally translated reading comprehension benchmark covering 122 language variants. Used as a complementary task performance benchmark alongside MKQA.